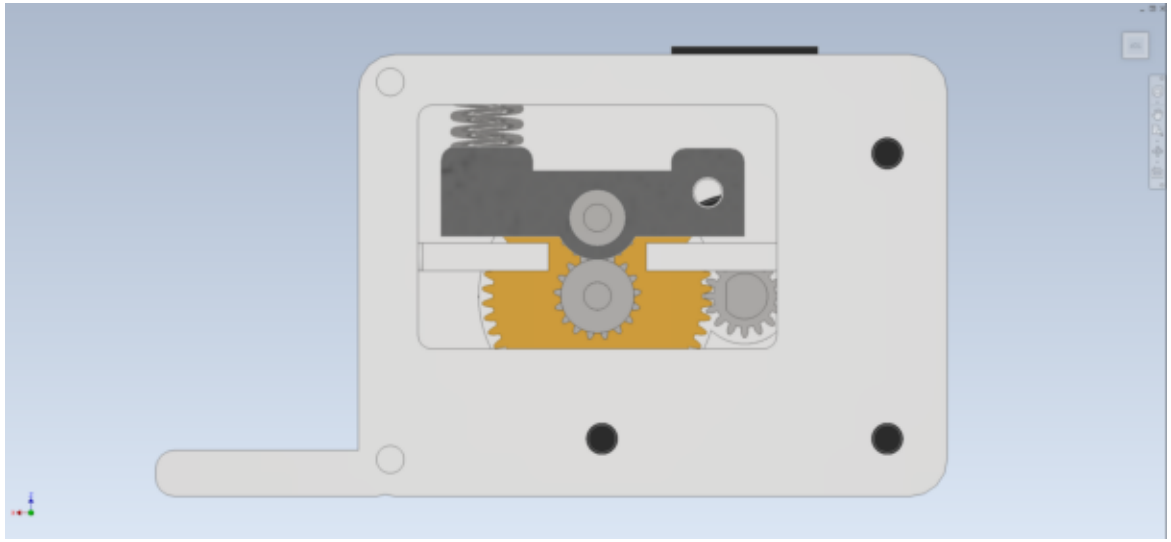
Filament Feeder: geared dual-drive Bowden feeder for 1.75 mm filament

FilamentFeeder - calculation report. Designed and calculated only: nothing has been printed, built or tested. CAD = Inventor, CALC = calculated, CAT = catalogue, ASSUMED.

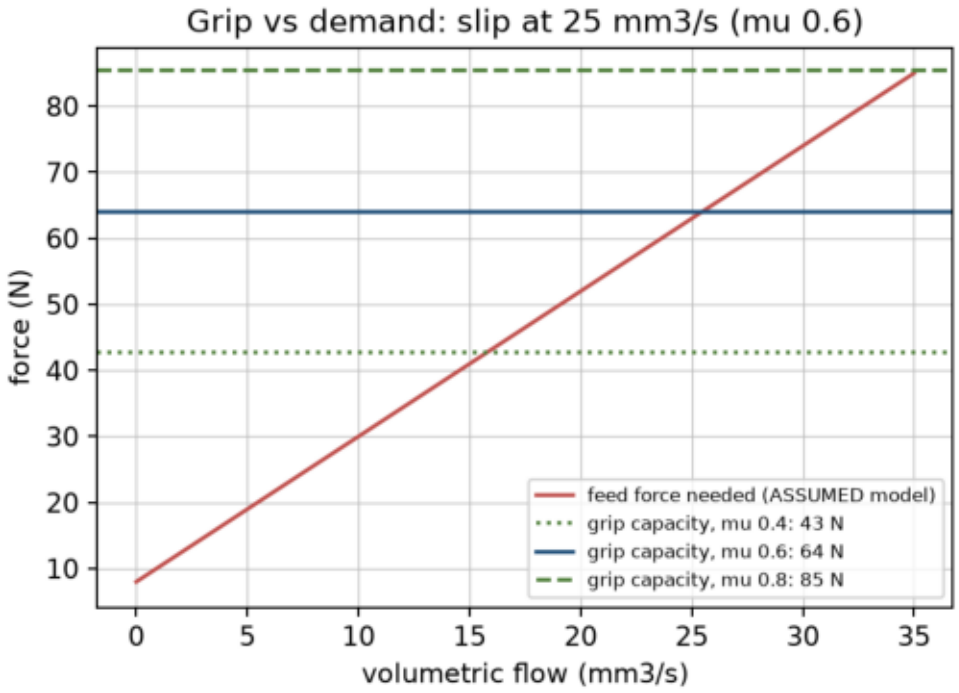


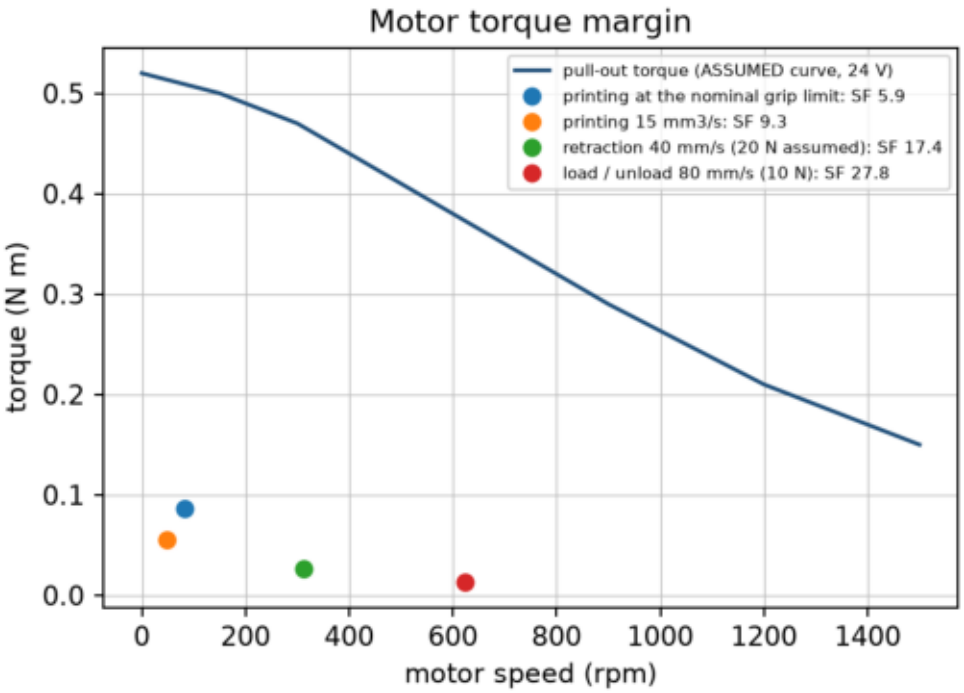
Item	Value	Source
Drive	NEMA 17 StepperOnline 17HS19-2004S1 (NEMA 17, 59 Ncm, 2.0 A, 48 mm); 16T / 48T m0.5 (3:1, brass 48T); two 17T dual-drive gears, d_eff 7.36 mm	CAT / CAD
Idler	PA6-CF arm on an M3 pivot, lever 2.0: spring 8 x 1 x 18 at 12 mm -> clamp 53.35 N	CALC
Path	PC4-M10 push-fits, 2.2 mm guides up to the gears, D2F runout switch block, PTFE 4 x 2 to the hot end	CAD
Spool	1 kg spool on a printed hub with 2 x 608ZZ, 8 mm axle in a printed upright, all on a 300 mm 2020 rail	CAD
Grip / flow	push capacity 64.0 N (mu 0.6), slip above about 25.5 mm3/s	CALC
Resolution	415.2 steps/mm (1/16)	CALC
Assembly	50 occurrences, 0 DOF, 0 clashes at 11 poses; 2.21 kg with a full spool	CAD

1. Grip, feed force and motor torque

FilamentFeeder - calculation report. Designed and calculated only: nothing has been printed, built or tested. CAD = Inventor, CALC = calculated, CAT = catalogue, ASSUMED.

Clamp: the spring (8 OD, 1 wire, 18 free, 6.5 active coils: k = G d^4 / 8 D^3 n = 4.446 N/mm) is installed at 12 mm, 26.68 N, and acts 24 mm from the pivot while the idler gear sits 12 mm from it: 53.35 N on the filament. Both drive gears drive (their 17T teeth mesh), so the push before slip is 2 mu N. mu for hardened teeth biting PLA is ASSUMED 0.4-0.8 (0.6 nominal): 42.7 / 64.0 / 85.4 N. The demand model (ASSUMED, PLA through a 0.4 nozzle at 215 C) is F_req = 8 N + 2.2 N per mm3/s (ASSUMED, PLA, 0.4 nozzle, 215 C): slip at 25.5 mm3/s nominal, 15.8 worst case. Motor: torque = F r_eff / (3 eta), eta 0.912 ASSUMED, against an ASSUMED 24 V pull-out curve for a 59 N cm 48 mm NEMA 17. Even at the grip limit the motor needs 0.086 N m at 82 rpm (SF 5.9): the motor is not the limit, the grip is. Resolution 415.2 steps/mm at 1/16 microstepping (2.41 um per microstep, 38.5 um per full step).



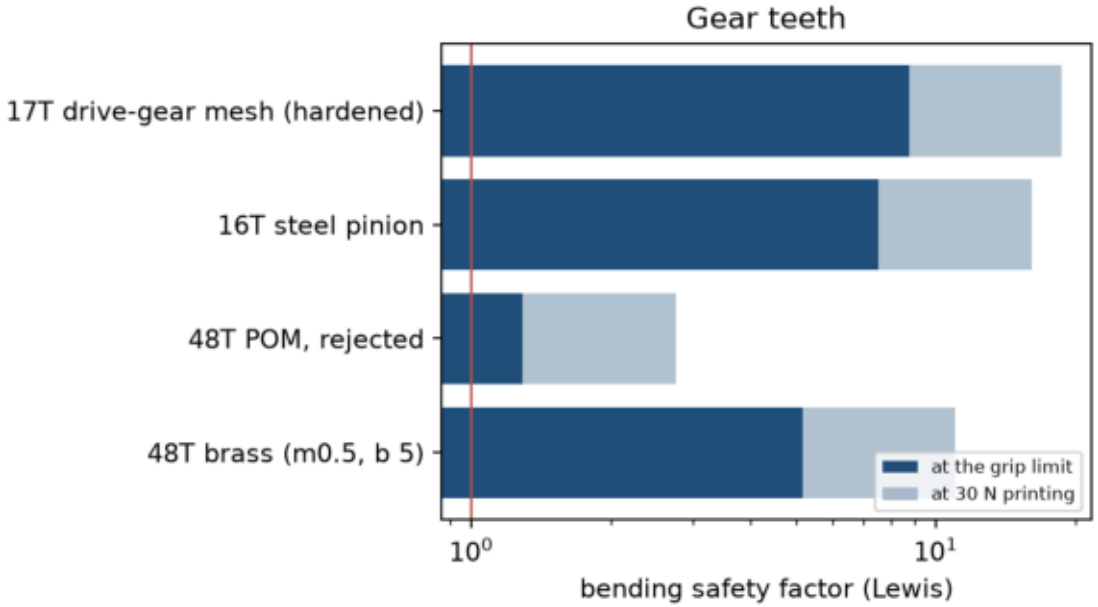


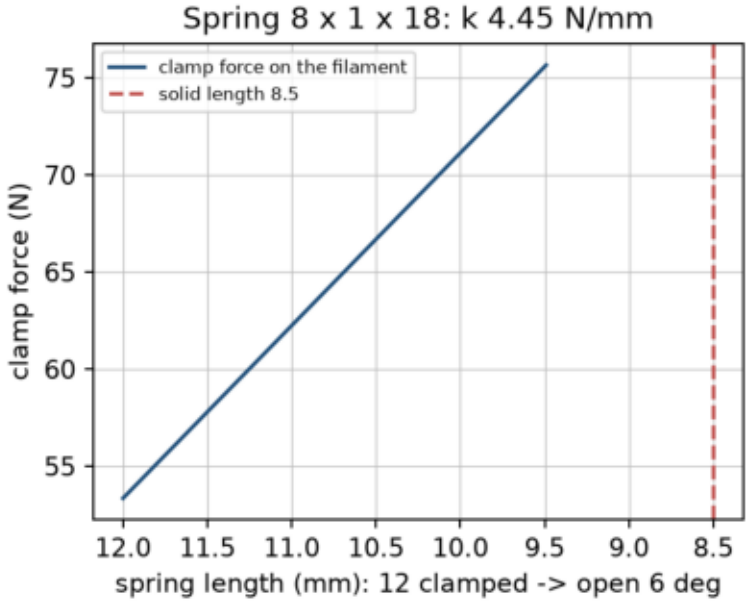
2. Gears, shafts, bearings, spring, printed parts

FilamentFeeder - calculation report. Designed and calculated only: nothing has been printed, built or tested. CAD = Inventor, CALC = calculated, CAT = catalogue, ASSUMED.

Part	Result	SF
48T brass (m0.5, b 5)	Ft 19.6 N, Y 0.406, 19.4 MPa vs 100	5.16 (11.02 at 30 N)
48T POM, rejected	Ft 19.6 N, Y 0.406, 19.4 MPa vs 25	1.29 (2.76 at 30 N)
16T steel pinion	Ft 19.6 N, Y 0.296, 26.5 MPa vs 200	7.54 (16.09 at 30 N)
17T drive-gear mesh (hardened)	Ft 27.7 N, Y 0.303, 45.7 MPa vs 400	8.75 (18.66 at 30 N)
Main shaft 3 mm, span 27	M 300 N mm + T 236 N mm: 137 MPa, 0.0187 mm at the groove	7.3
Idler shaft 3 mm	71 MPa, 0.0038 mm	14.1
Spring	577 / 818 MPa clamped / open vs 0.45 Su = 969; bind margin 0.99 mm	1.68 / 1.18
Idler arm (PA6-CF)	5.6 MPa at the bearing seat (net section)	7.98
Spool upright (PETG)	1.23 MPa across layers, axle 9.8 MPa	12.2

Design change from these numbers: the 48T gear was POM (Lewis SF 1.29 at the grip-limited force, where a slipping filament loads the gear hardest) and is now brass (SF 5.2). The idler arm is PA6-CF: strength passes in PETG too, but the spring load is constant, so creep decides.





3. Test summary

FilamentFeeder - calculation report. Designed and calculated only: nothing has been printed, built or tested. CAD = Inventor, CALC = calculated, CAT = catalogue, ASSUMED.

Test	Result	Criterion	Status
Working model (Inventor)	50 occurrences, 147 constraints, 0 DOF; Feed and Lever drive every gear, the motor shaft, spool, filament and idler arm; 0 clashing pairs at 11 poses; probes within 0.00000 deg	0 DOF, 0 clashes	PASS
Gear meshes (involute, CAD)	16/48 and 17/17 clash-free at every pose; pinion backlash 0.0566 mm; drive gears in flank contact	no overlap	PASS
Clearances (Inventor)	filament to grooves 0.0669 mm (envelope), loading gap 0.8991 mm at 6 deg, motor shaft to guide 0.2612 mm	> 0	PASS
Clamp and grip	spring 8 x 1 x 18, k 4.446 N/mm, 26.68 N x lever 2.0 = 53.35 N; push capacity 42.7-85.4 N (mu 0.4-0.8 ASSUMED), 64.0 N nominal	>= 50 N	PASS (mu to be measured)
Flow before slip	25.5 mm3/s nominal (15.8-35.2); model F_req = 8 N + 2.2 N per mm3/s (ASSUMED, PLA, 0.4 nozzle, 215 C)	>= 15 mm3/s	PASS (assumed demand model)
Motor torque margin	printing at the nominal grip limit: SF 5.9; printing 15 mm3/s: SF 9.3; retraction 40 mm/s (20 N assumed): SF 17.4; load / unload 80 mm/s (10 N): SF 27.8	>= 2	PASS (pull-out curve assumed)
Resolution	415.2 steps/mm at 1/16 (2.41 um / microstep)	report	-
Gear teeth (Lewis, grip-limited)	48T brass (m0.5, b 5): SF 5.16; 48T POM, rejected: SF 1.29; 16T steel pinion: SF 7.54; 17T drive-gear mesh (hardened): SF 8.75	>= 2	PASS (brass 48T; POM rejected)
Shafts	main 3 mm: 137 MPa (SF 7.3), deflection 0.0187 mm; idler 3 mm: 71 MPa	SF >= 2	PASS
Bearings	683ZZ [15.8, 37.5] N, MR63ZZ 26.7 N: L10 > 10^6 h at 15 mm3/s	report	PASS
Spring	stress 577 MPa clamped (SF 1.68), 818 MPa open (SF 1.18); coil-bind margin 0.99 mm	SF >= 1.1, > 0	PASS
Idler arm (PA6-CF)	5.6 MPa at the bearing seat (SF 7.98); creep decides the material	>= 2	PASS
Spool upright + axle	post 1.23 MPa across layers (SF 12.2), axle 9.8 MPa	>= 3	PASS
Spool drag	0.035 N steady, 2.01 N at 2000 mm/s2	<< grip	PASS
Printability	FeederBody: 4.1 % overhang; FeederCover: 0.5 % overhang; RunoutBody: 1.1 % overhang; RunoutLid: 0.0 % overhang; SpoolUpright: 0.4 % overhang; SpoolHub: 4.8 % overhang; IdlerArm: 5.3 % overhang	report	PASS (supports only in the body gear pocket)